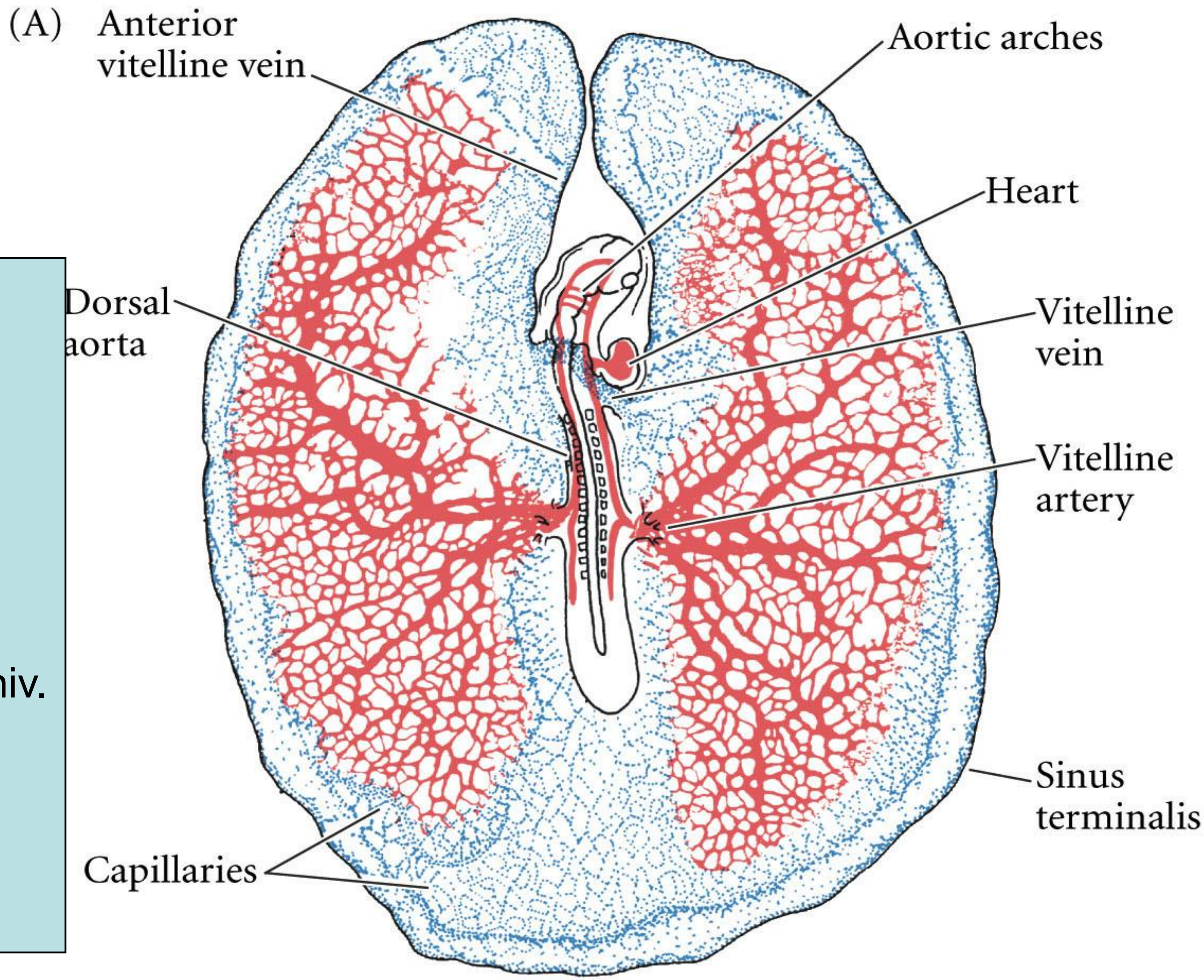
Blood vessel development



Nandor Nagy

Semmelweis Univ.

Vasculogenesis: de novo blood vessel generation from the lateral plate mesoderm

Angiogenesis: remodeling and pruning into capillary bed, arteries, and veins

Lymphangigenesis

Vasculogenesis

1. First phase

Initiated from the generation of hemangioblasts, which leave the primitive streak in the posterior region of the embryo; a part of splanchnic mesoderm

Blood islands: condensed cell mass derived from the splanchinc mesoderm

inner cells = blood progenitor cells; outer cells = angioblasts

2. Second phase

Angioblasts proliferate and differentiate into endothelial cells

3. Third phase

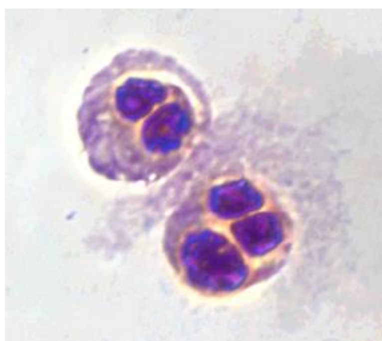
Endotheilial cells form primary capillary plexus

Formation of vessels by differentiation of cells from angioblasts in the yolk sac of the embryo:

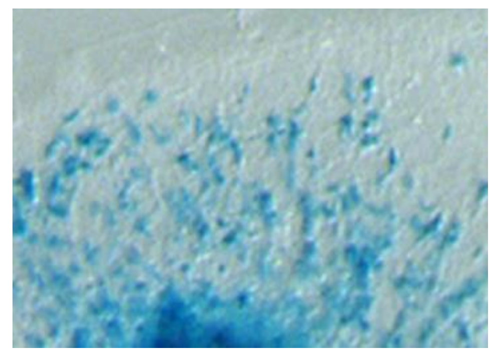
Leads to formation of a primitive tubular network in a non-vascularized tissue

Hematopoiesis and vasculogenesis closely linked -- spatially and temporally -- during embryonic development

The hemangioblast

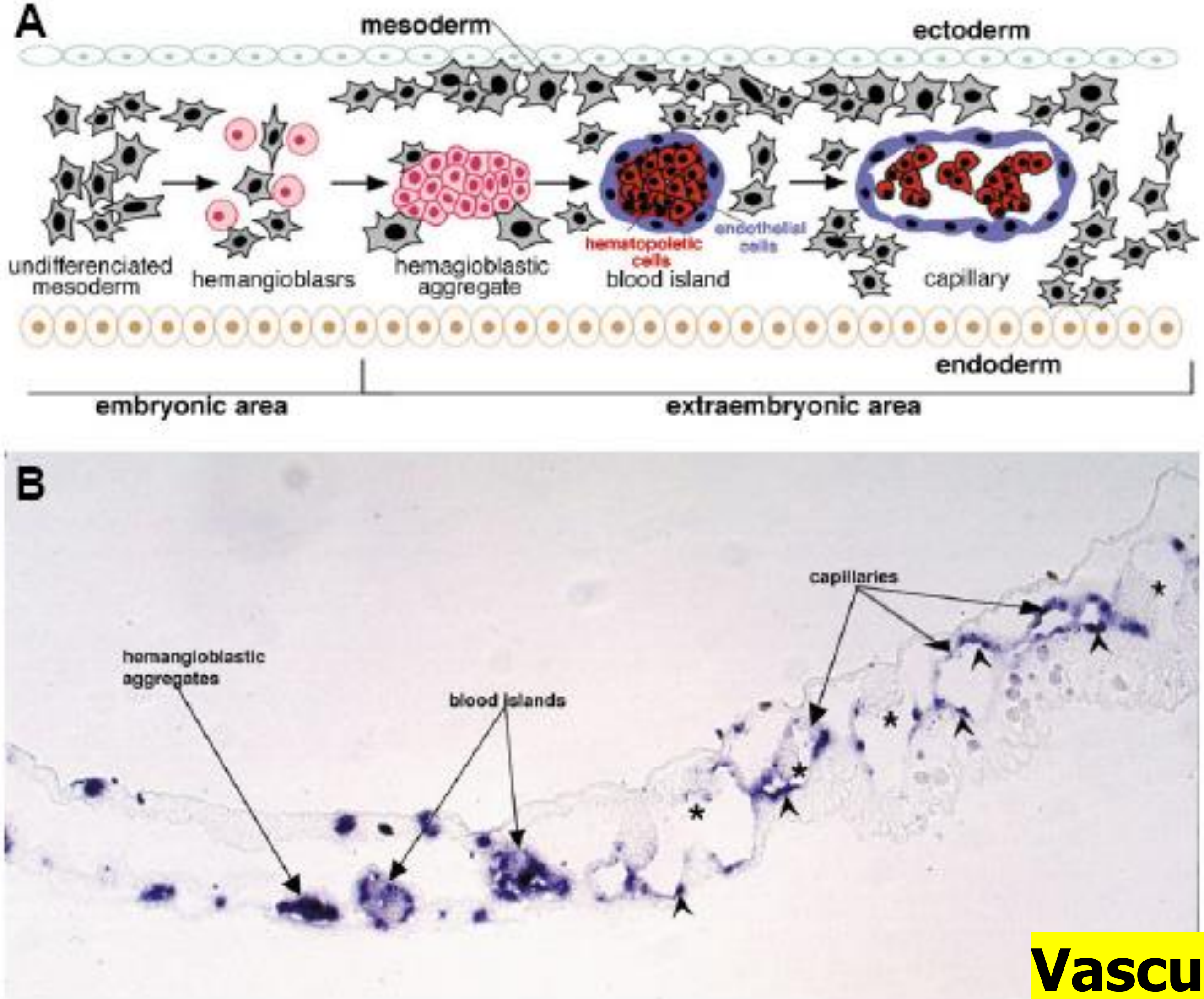


hematopoietic



endothelial

Hypothetical common progenitor for hematopoietic and vascular endothelial cells

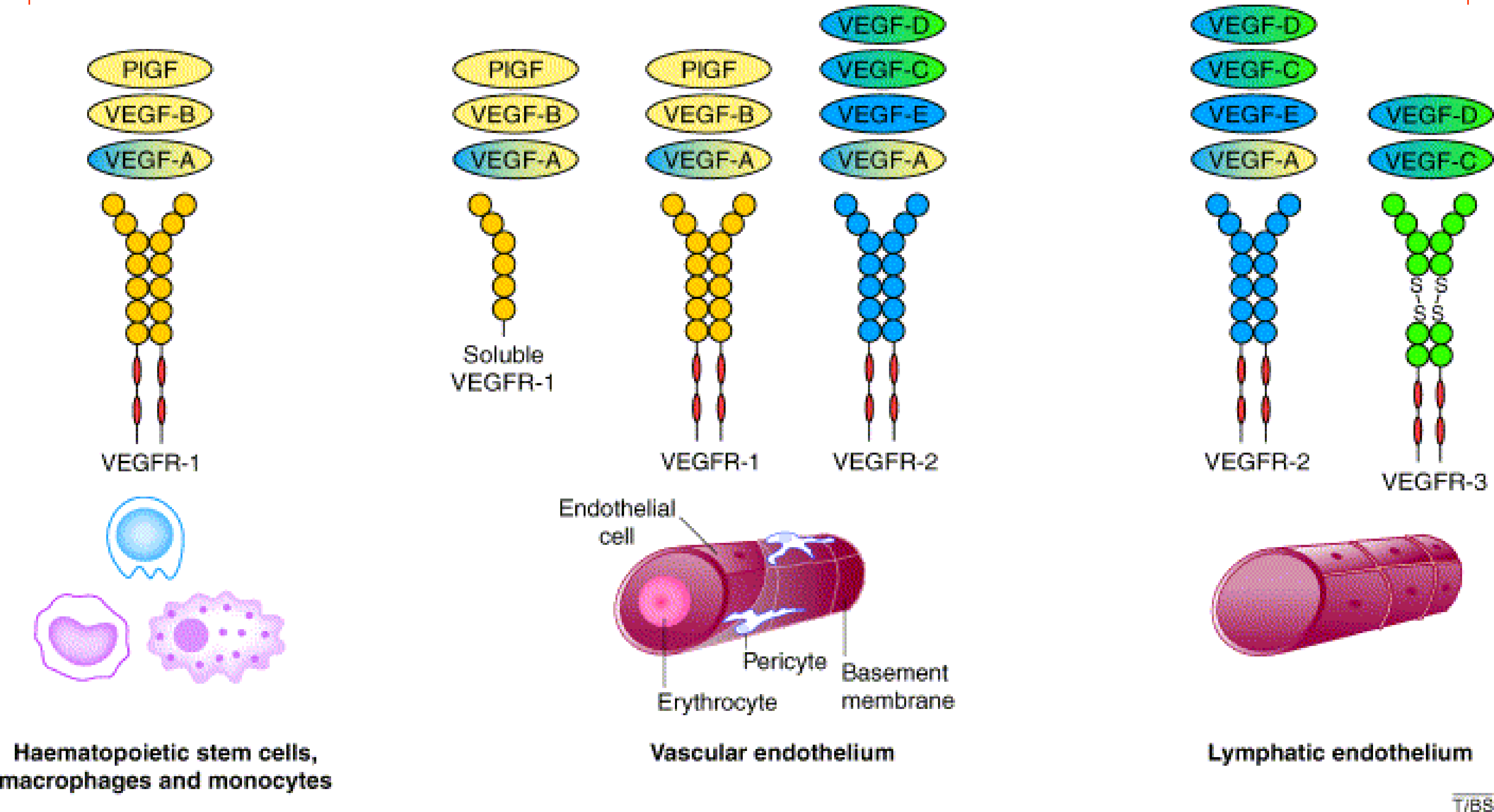


Vasculogenesis

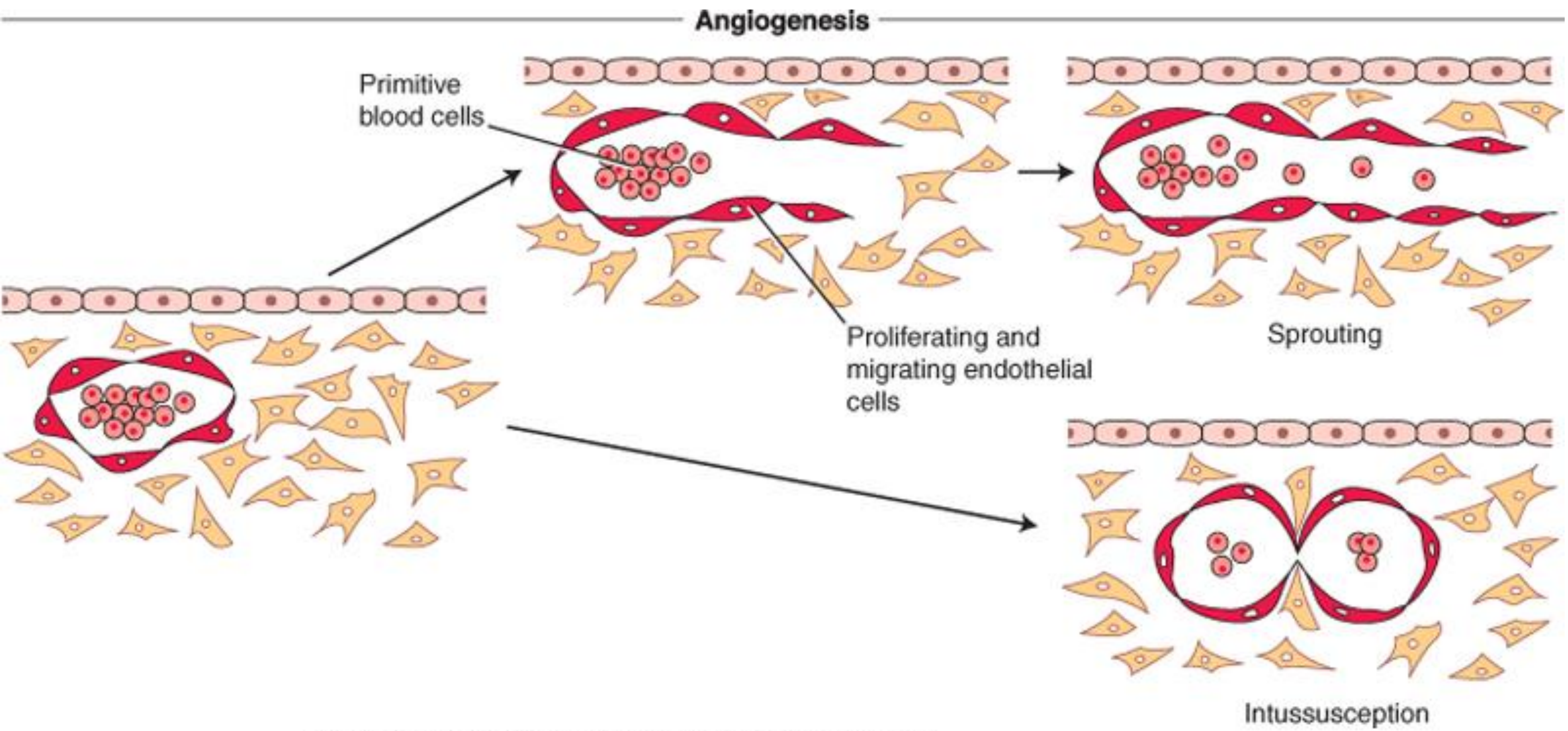
Fig. 3. Differentiation of yolk sac blood islands. (A) *Schematic representation: see text for details.* **(B)** *Transverse section through the extraembryonic area of a 15-somite stage embryo, labeled with an antisense riboprobe specific for VEGFR2. Hemangioblastic aggregates are positive. In the blood islands and capillaries, expression of the receptor decreases in the hematopoietic cells (asterisks), but is maintained in the endothelial cells (arrowheads). Bar, 75 μm.*

VEGF (Vascular Endothelial Growth Factor)

Discovered in 1989. Role: endothelial cell proliferation, differentiation, migration and proper vessel patterning. A heparin binding growth factor released by many tissues. Basis of tumor/tissue neovascularization. Binds to tyrosine kinase active receptor. HIF-1a inducible



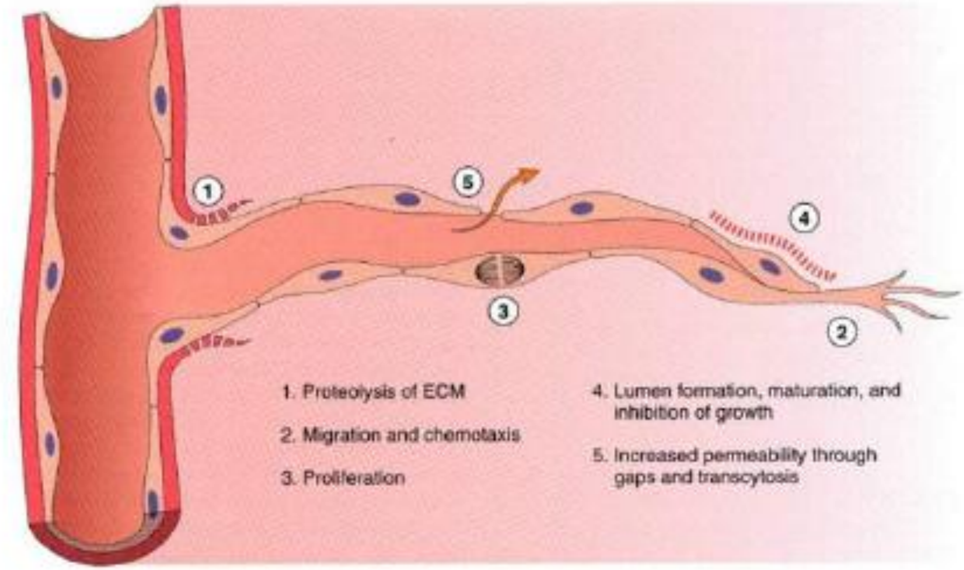
Angiogenesis: sprouting and intusescepion

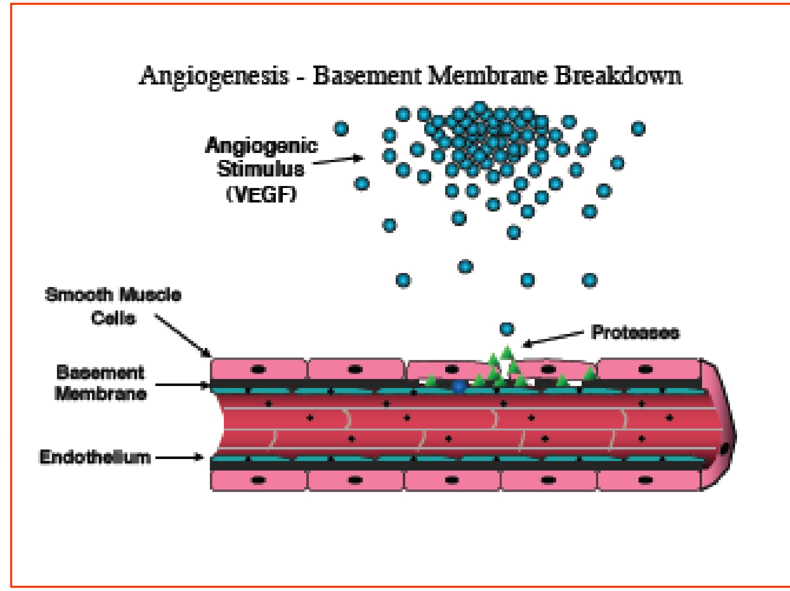


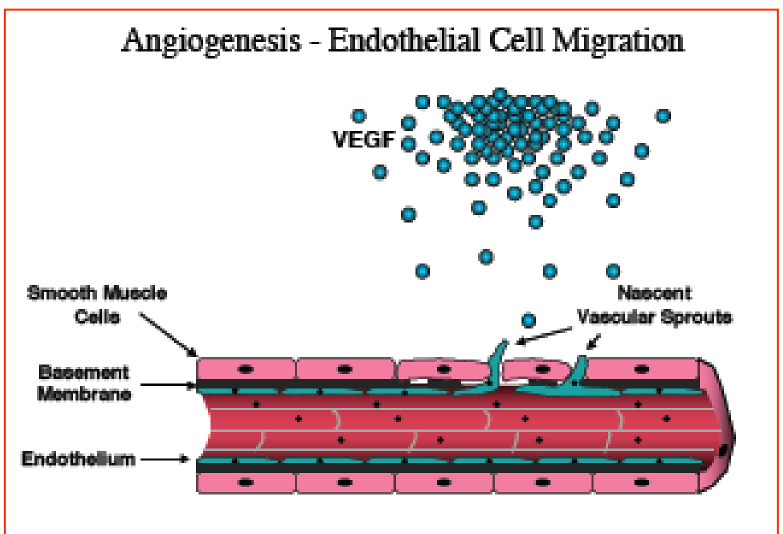
Schoenwolf et al: Larsen's Human Embryology, 4th Edition.

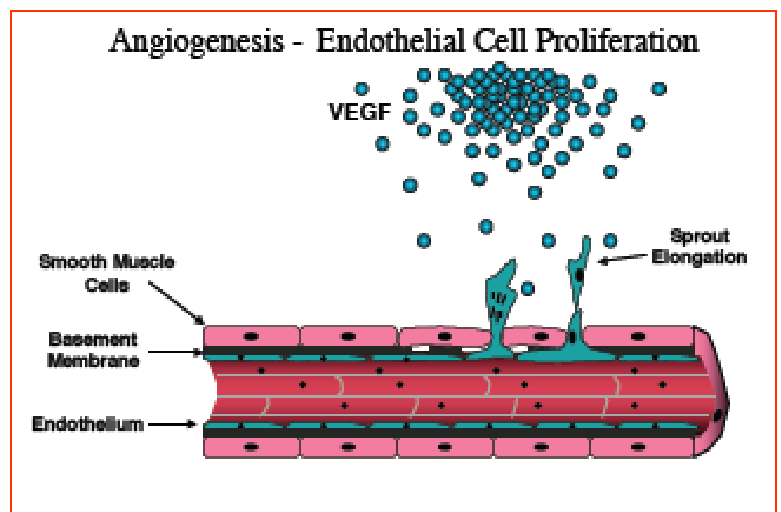
Angiogenesis

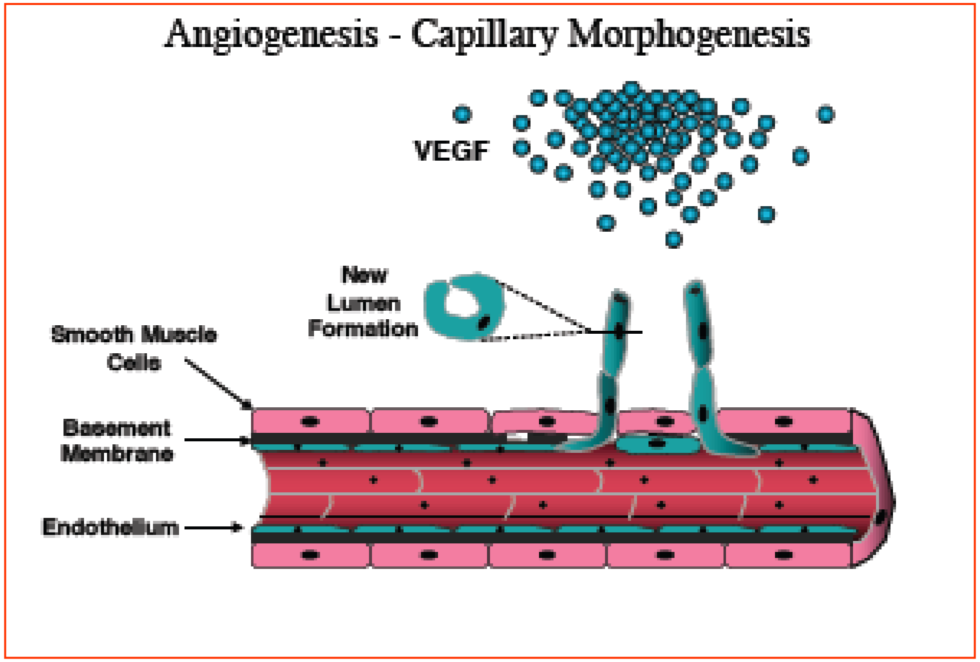
Sprouting of cells from mature endothelial cells froma preexisting vessel wall



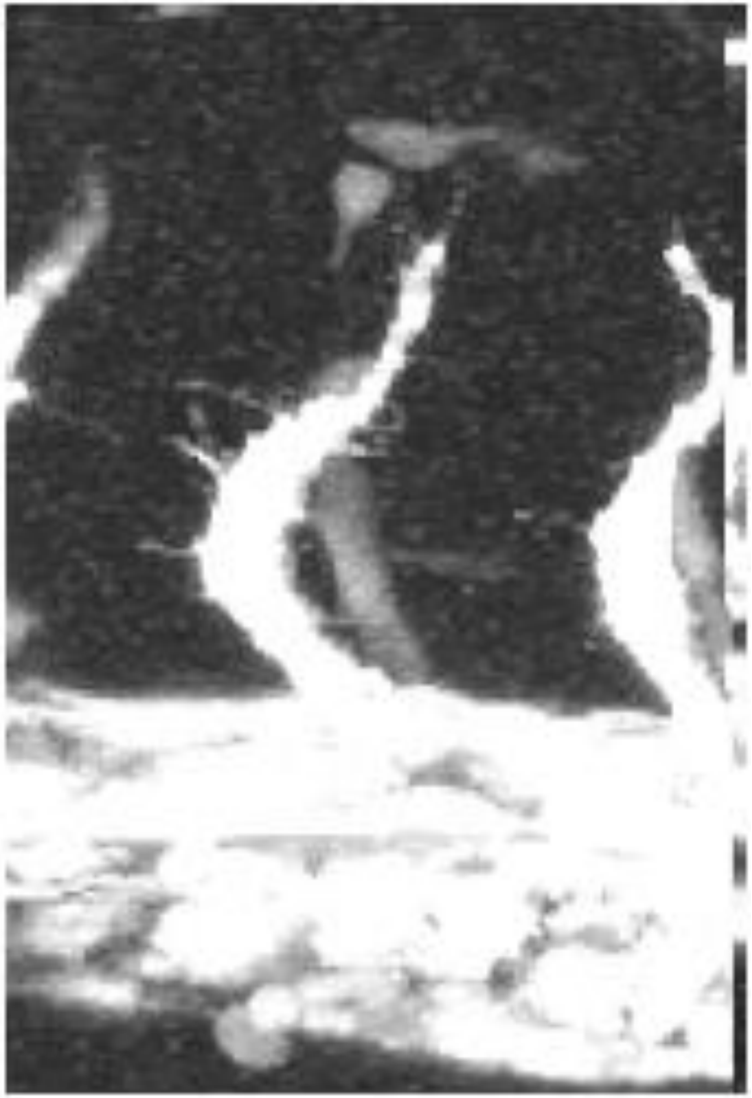


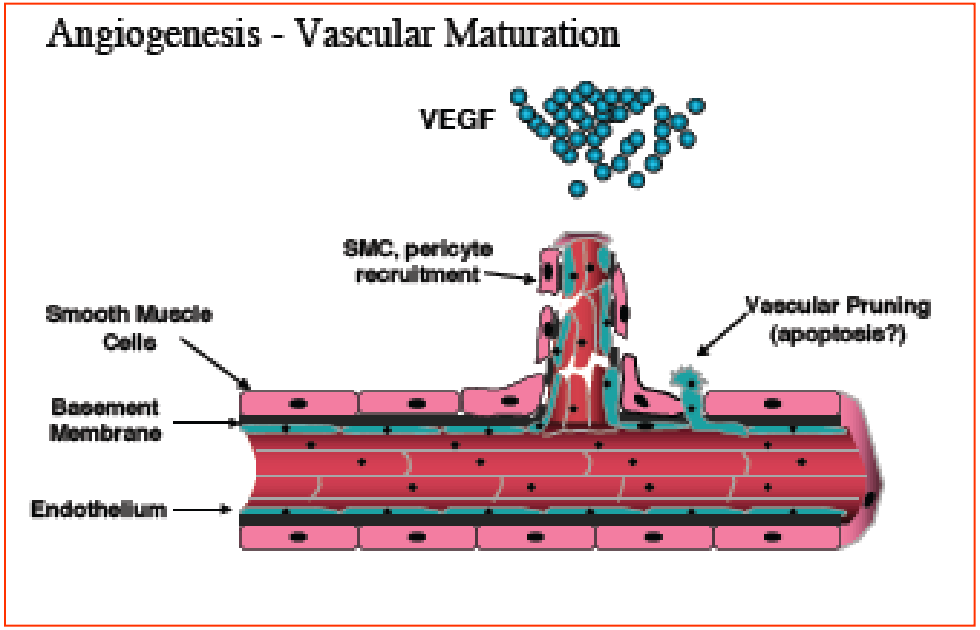





Angiogenesis - Capillary Morphogenesis
VEGF
New
Lumen
Formation
Smooth Muscle
Cells
Basement
Membrane
Endothelium

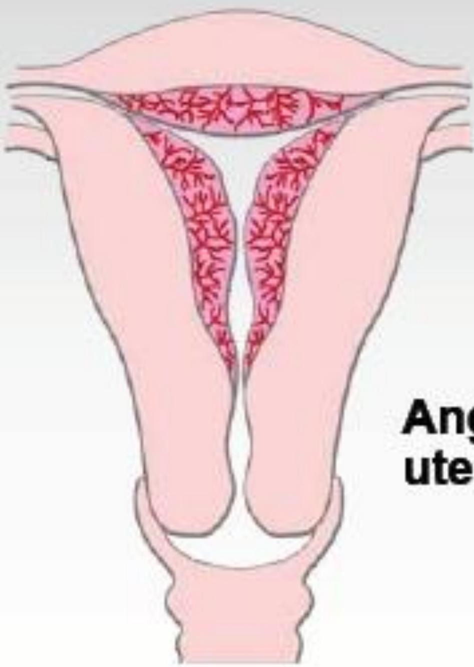
Sprouting !




Angiogenesis - Vascular Maturation
VEGF
SMC, pericyte
recruitment
Smooth Muscle
Cells
Vascular Pruning
(apoptosis?)
Basement
Membrane
Endothelium

Angiogenesis occurs in normal adult tissues such as:

- Ovaries (corpus luteum)
- In teeth (especially in rodents)
- During wound healing



Angiogenesis in uterine lining

Angiogenesis in tissue during wound healing

At 4 weeks age embryo:

- 2 chambered heart
- a blood vascular system which consists of 3 separate circulatory arcs

intraembryonic circulatory arc
(is organized in a similar manner to that of a fish)

vitelline or omphalomesentheric
arc (from and to the yolk sac) extraembyonic

umbilical vessles
(body stalk: chorion, primitive placenta), extrembryonic

Basic circulatory arches in the 4-week old human embryo

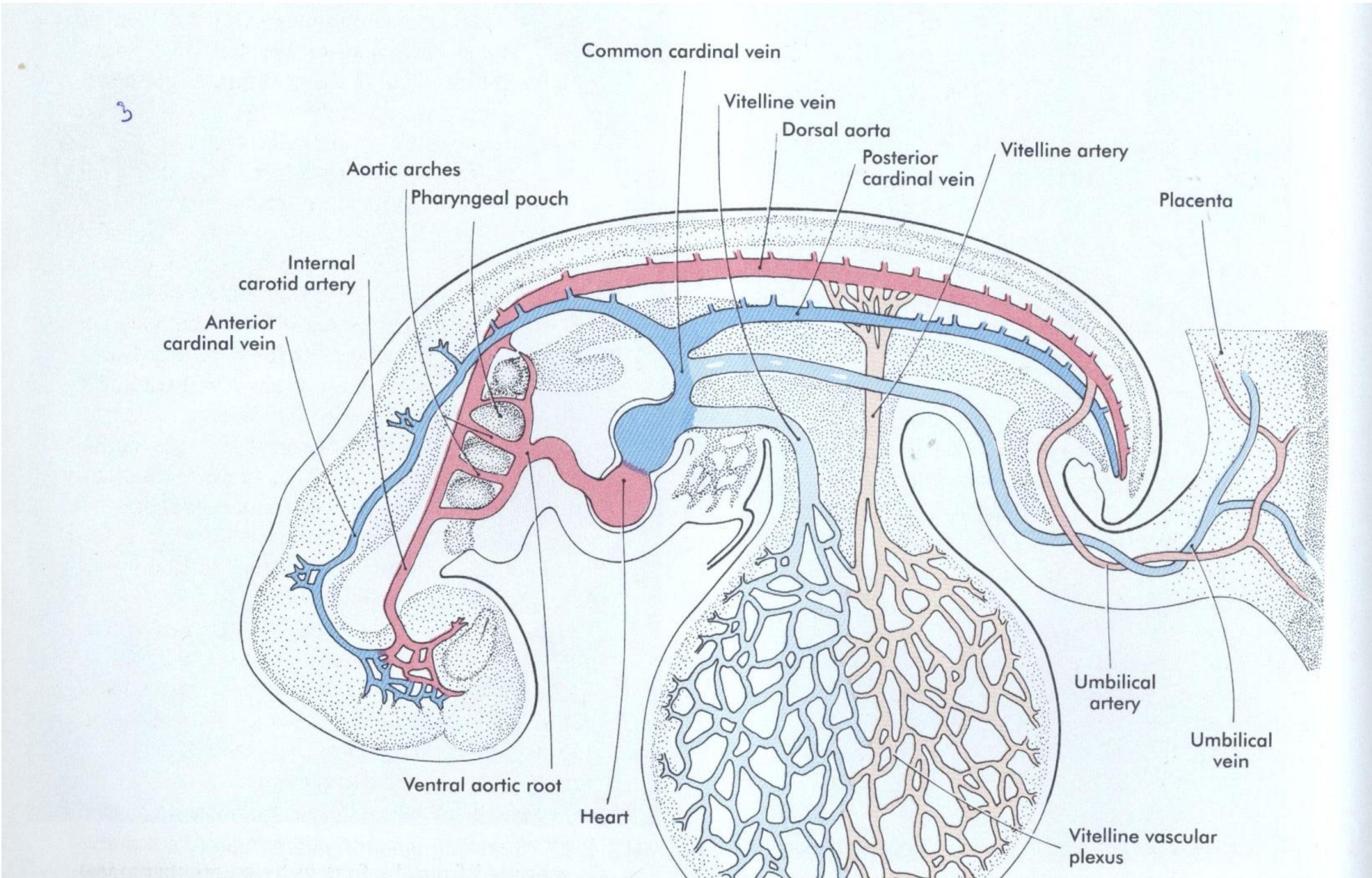
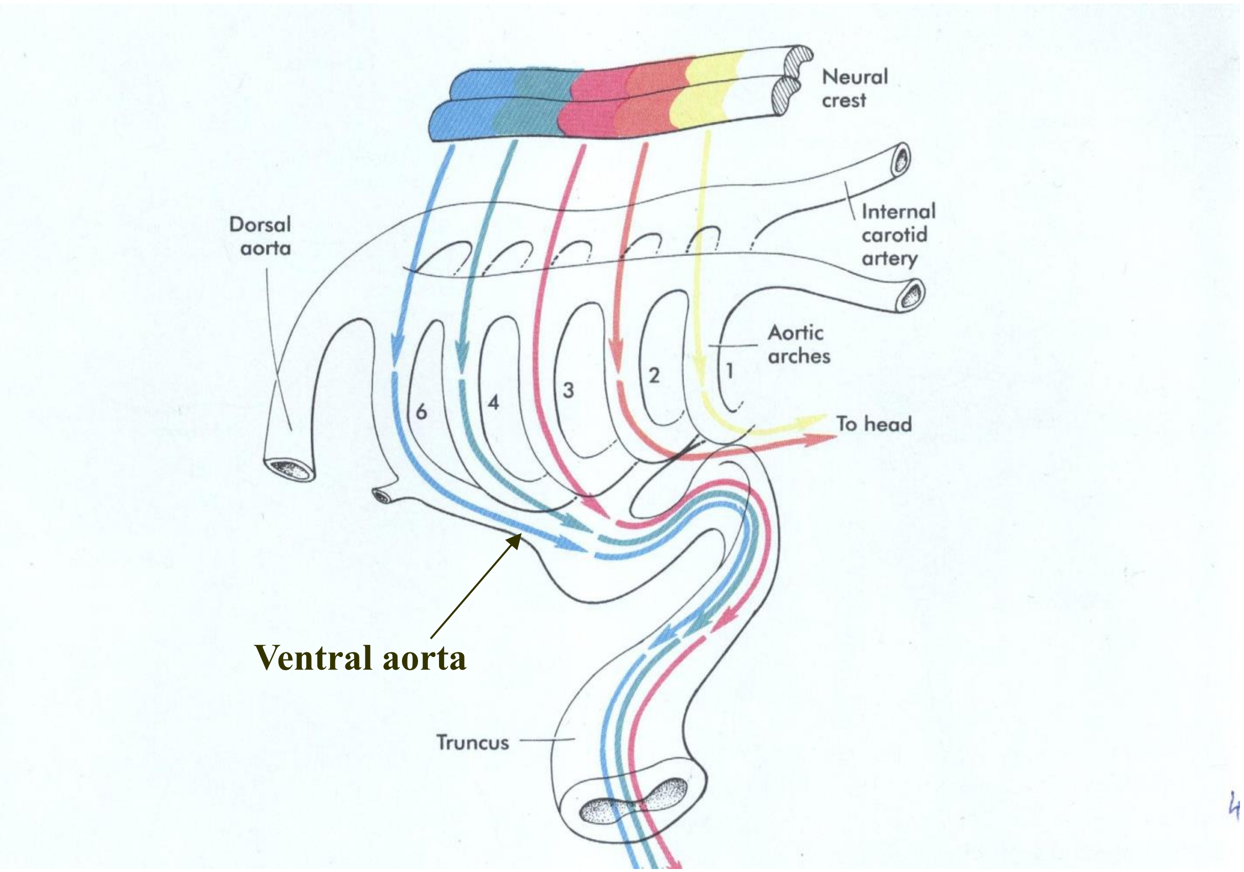


FIG. 5-25 Basic circulatory arcs in the 4-week-old human embryo.

Intraembryonic circulatory arch



common ventricle

↓

ventral aortic root
(truncus arteriosus)

↓

2 primitive aorta ascendens

↓

6 pairs aortic arches
(through the branchial arches)

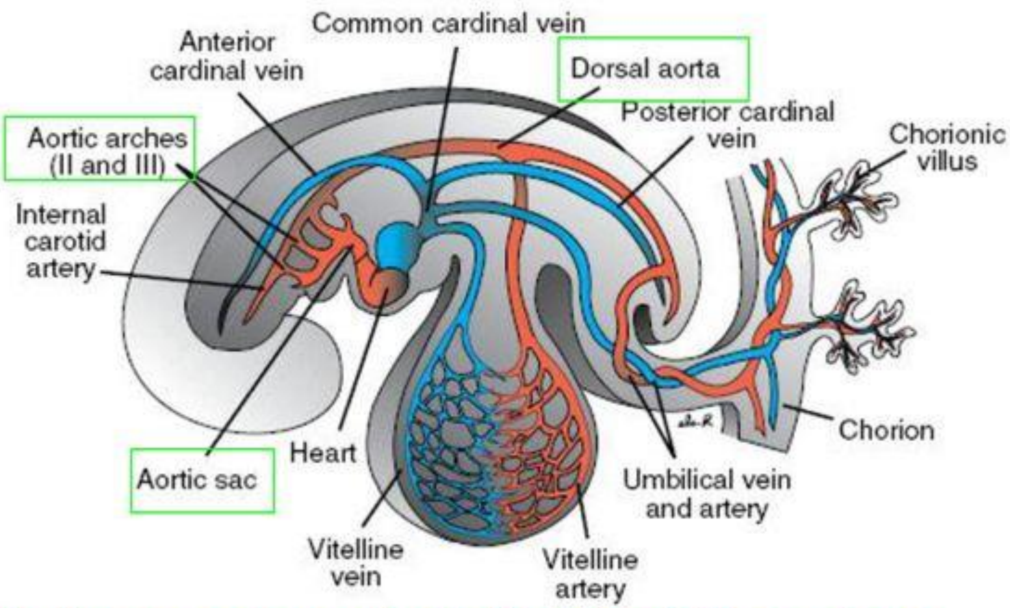
↓

paired dorsal aortae

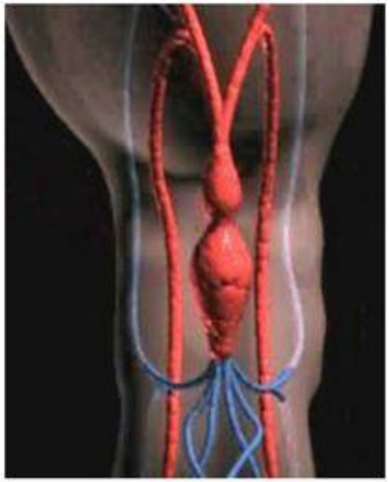
↓

unpaired dorsal aorta

- The aortic arches terminate in right and left dorsal aortae. (In the region of the arches the dorsal aortae remain paired, but caudal to this region they fuse to form a single vessel.)



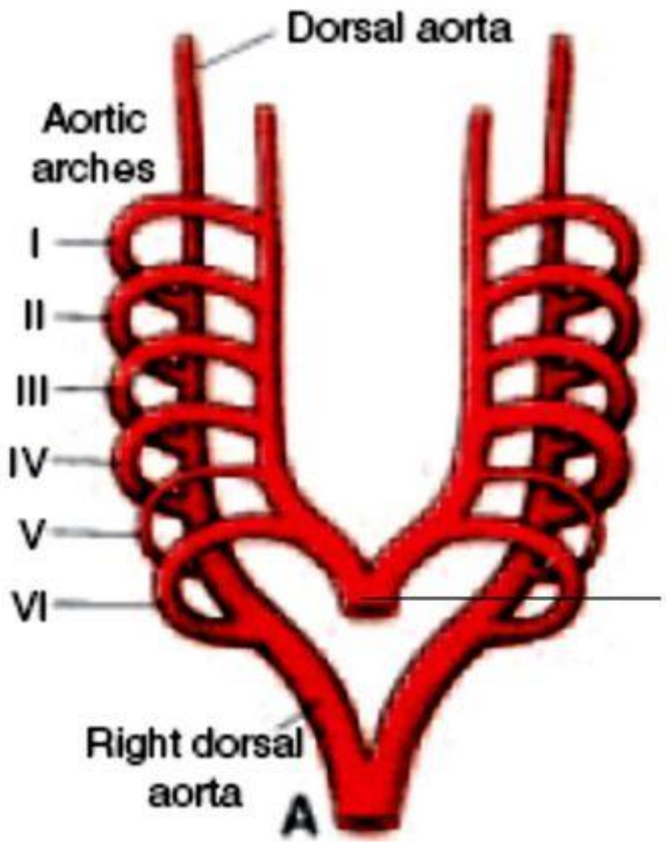
Only the vessels on the left side of the embryo are shown.



Ventral view of the embryo

- The aortic aches appear in a cranial to caudal sequence gradually.

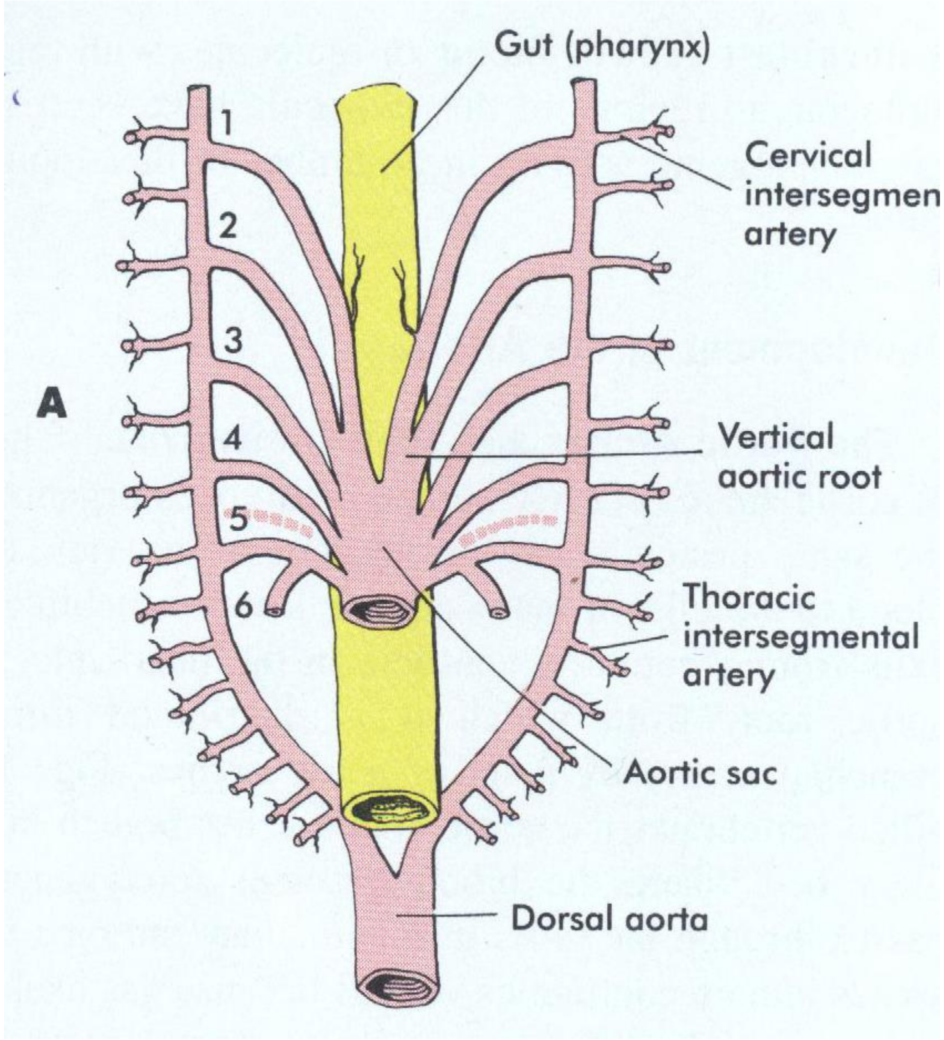
- The aortic sac gives rise to a total of **six pairs** of arteries. During further development, some vessels regress completely.



The fifth pair is rudimentary and disappears at a very early stage

A. Aortic arches and dorsal aortae before transformation into the definitive vascular pattern.

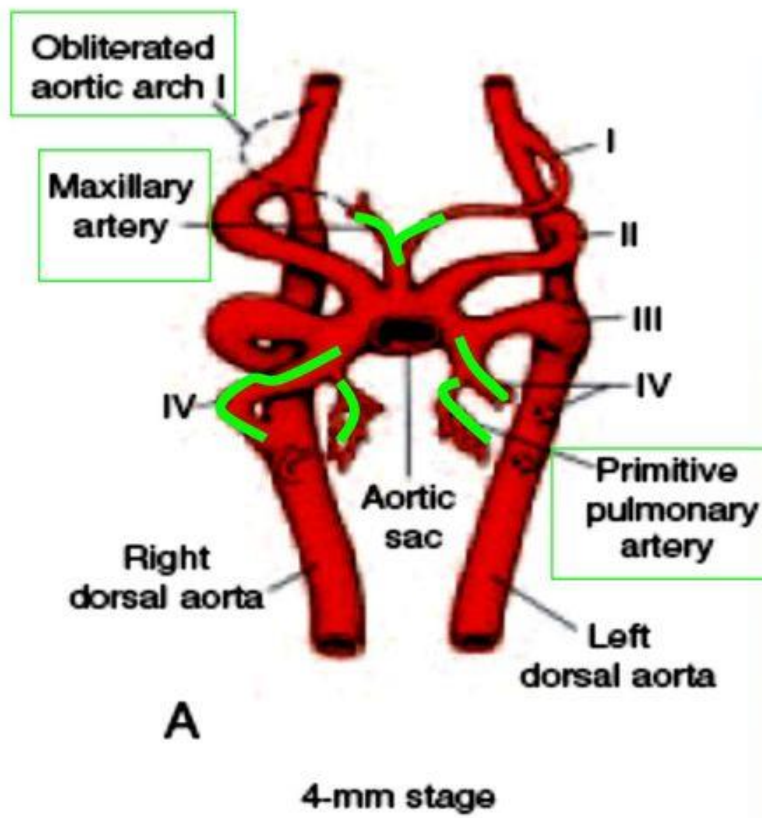
Branchial (pharyngeal) aortic arches



In human embryo all the aortic arches are NEVER present at the same time!!!!

Arch I:

- By day 27, most of **1st aortic arch** has disappeared on both sides, a small portion persists to form **maxillary artery.**



- Arch II:
- 2nd aortic arch soon disappears.
- The remaining portions are **hyoid** and **stapedial arteries.**

- Arch III: CAROTID ARCH – Persists becomes part of carotid arteries.
- 1- Common carotid artery
- 2-Proximal part of internal carotid artery
- 3-External carotid artery

Arch IV: AORTIC ARCH
Right side: Rt subclavian -
Left side : Main Part of the ARCH OF AORTA.

Arch V: **DISAPPEARS**

Arch VI: PULMONARY ARCH –
On the left side: **Ventral part**: Left pulmonary artery
Dorsal part: Ductus arteriosus

On the rt side: **Ventral part**: Right Pulmonary artery
Dorsal part: Disappears

(No ductus arteriosus on the right side)

Arch of Aorta develops from 3 parts:

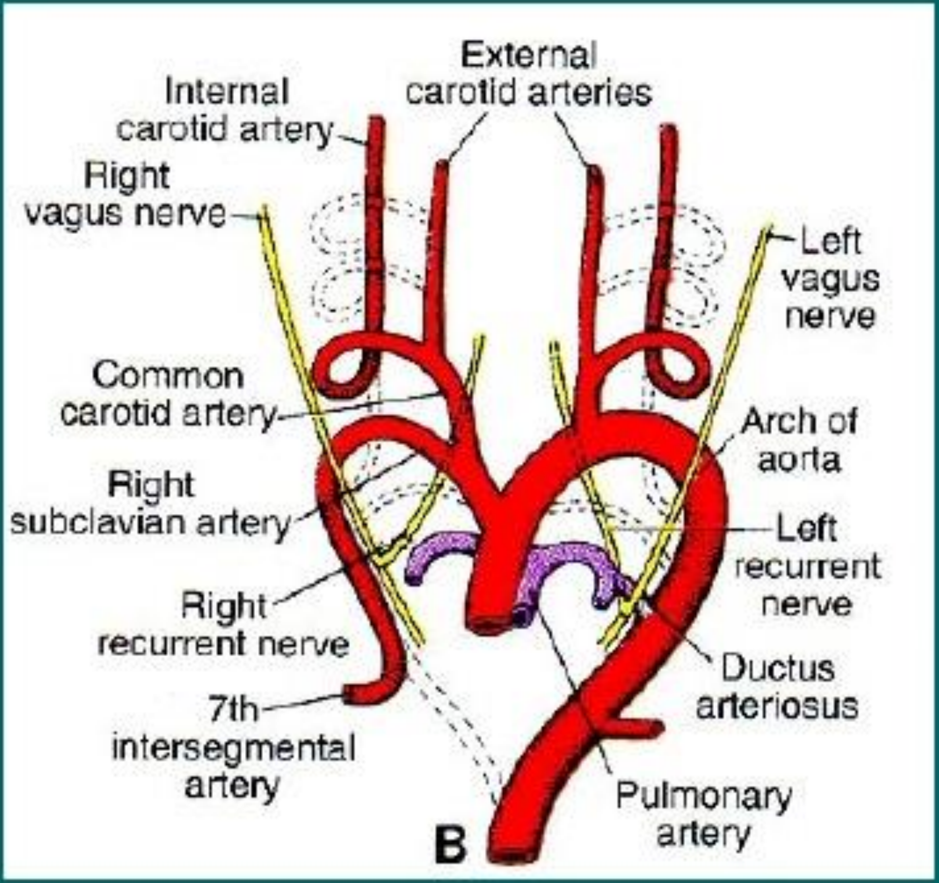
- 1) Proximal part : from the left part of the aortic sac
- 2) Middle part: from the left 4th aortic arch
- 3) Distal Part: From the dorsal aorta between the left fourth and 6th arches

the 1st aortic arch – **disappears** (a small portion persists and forms a piece of the maxillary artery)

the 2nd aortic arch – **disappears** (small portions of this arch contributes to the hyoid and stapedial arteries)

the 3rd aortic arch - has the same development on the right and left side

it gives rise to the initial portion of **the internal carotid artery**, the remainder of its trunk is formed by the cranial portion of the dorsal aorta + primitive internal carotid



the external carotid is deriving from the cranial portion of the ventral aorta

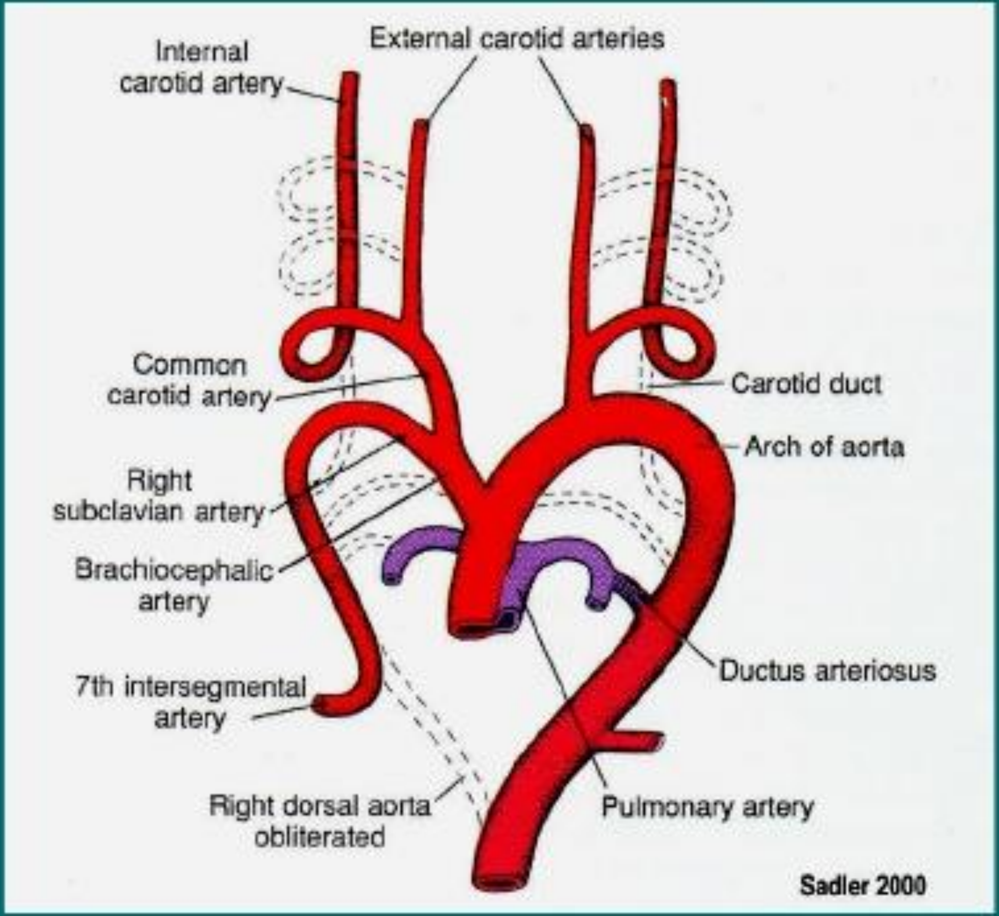
the common carotid corresponds to a portion of the ventral aorta between exits of the third and fourth arches

the 4th aortic arch - has ultimate fate different on the right and left side

on the left - it forms **a part of the arch of the aorta** between left common carotid and left subclavian artery

on the right - it forms the proximal segment **of the right subclavian artery**

the 5th aortic arch - is transient and soon obliterates

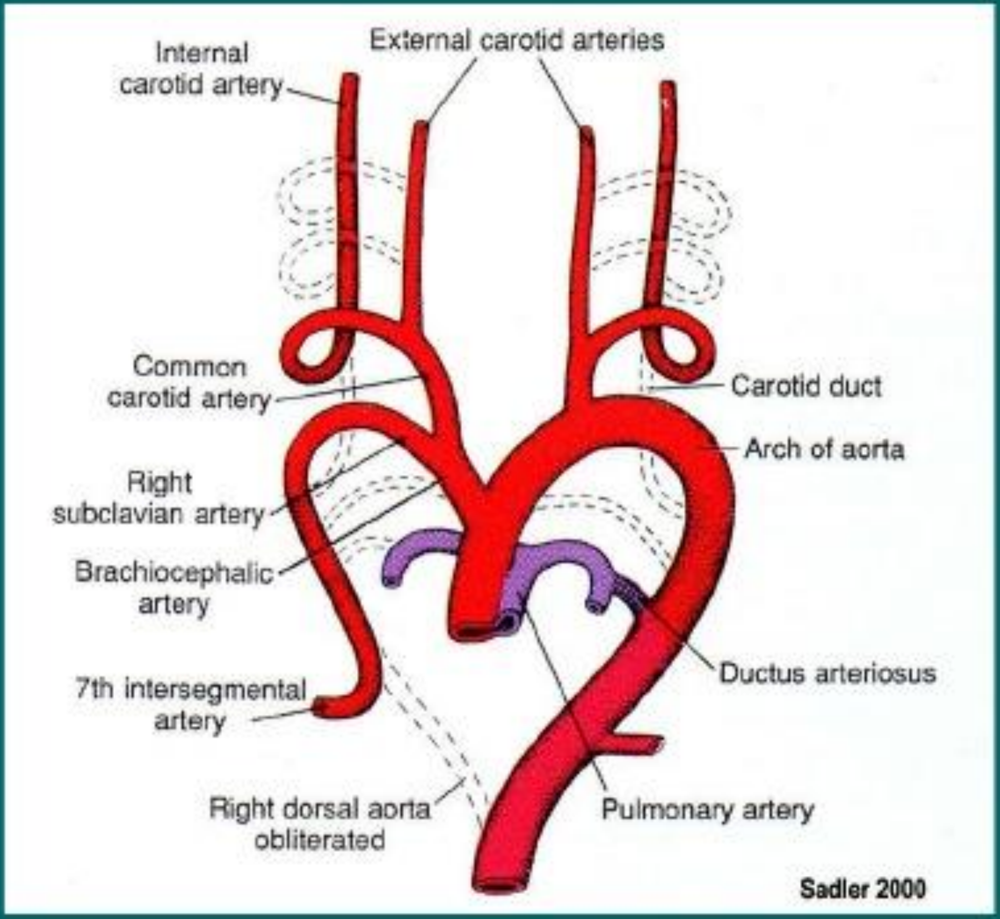


the 6th aortic arch - **pulmonary arch** - gives off a branch on each side that grows toward the developing lung bud

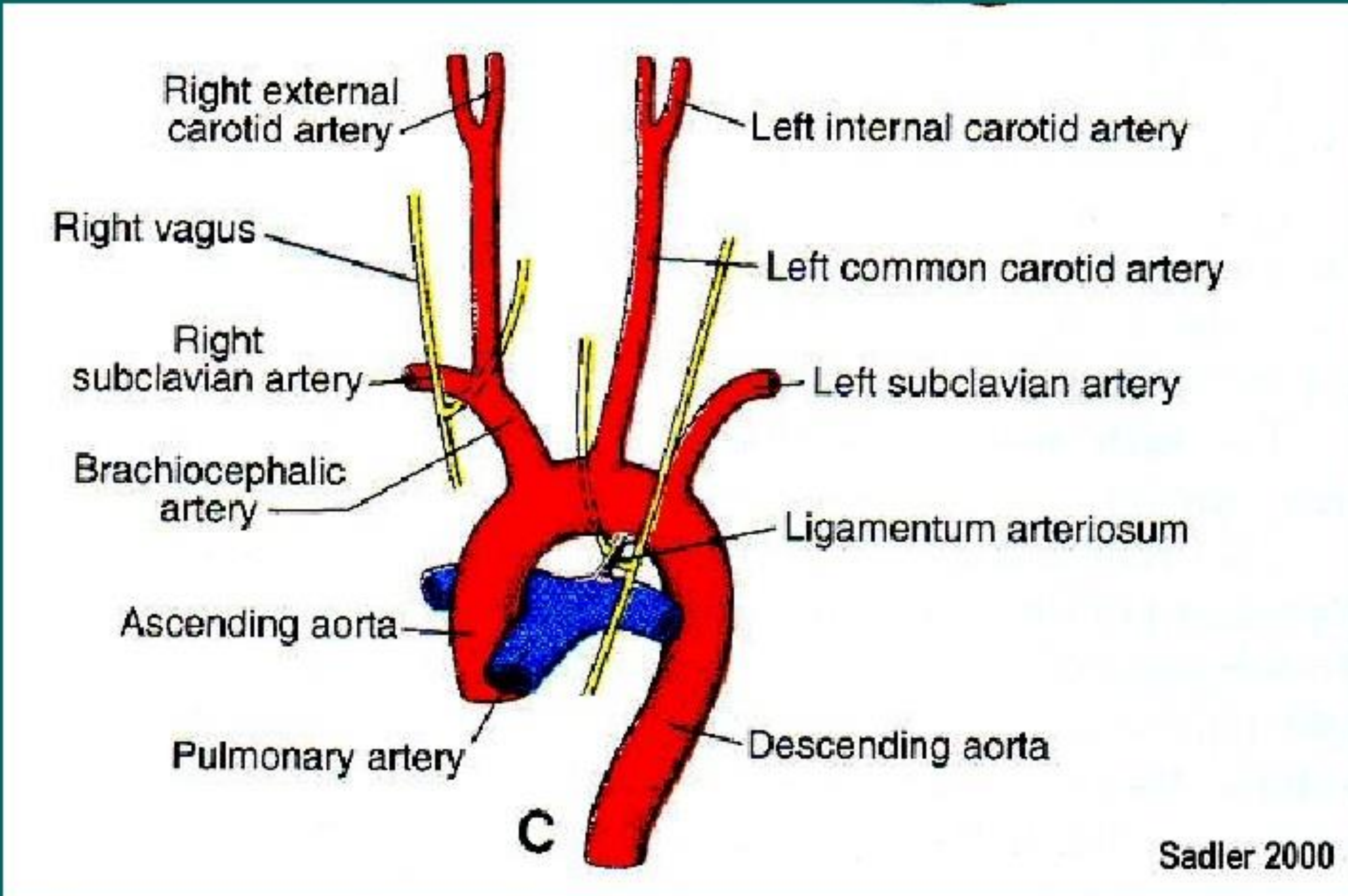
on the right side, the proximal part transforms into the right branch of the pulmonary artery and the distal part disappears

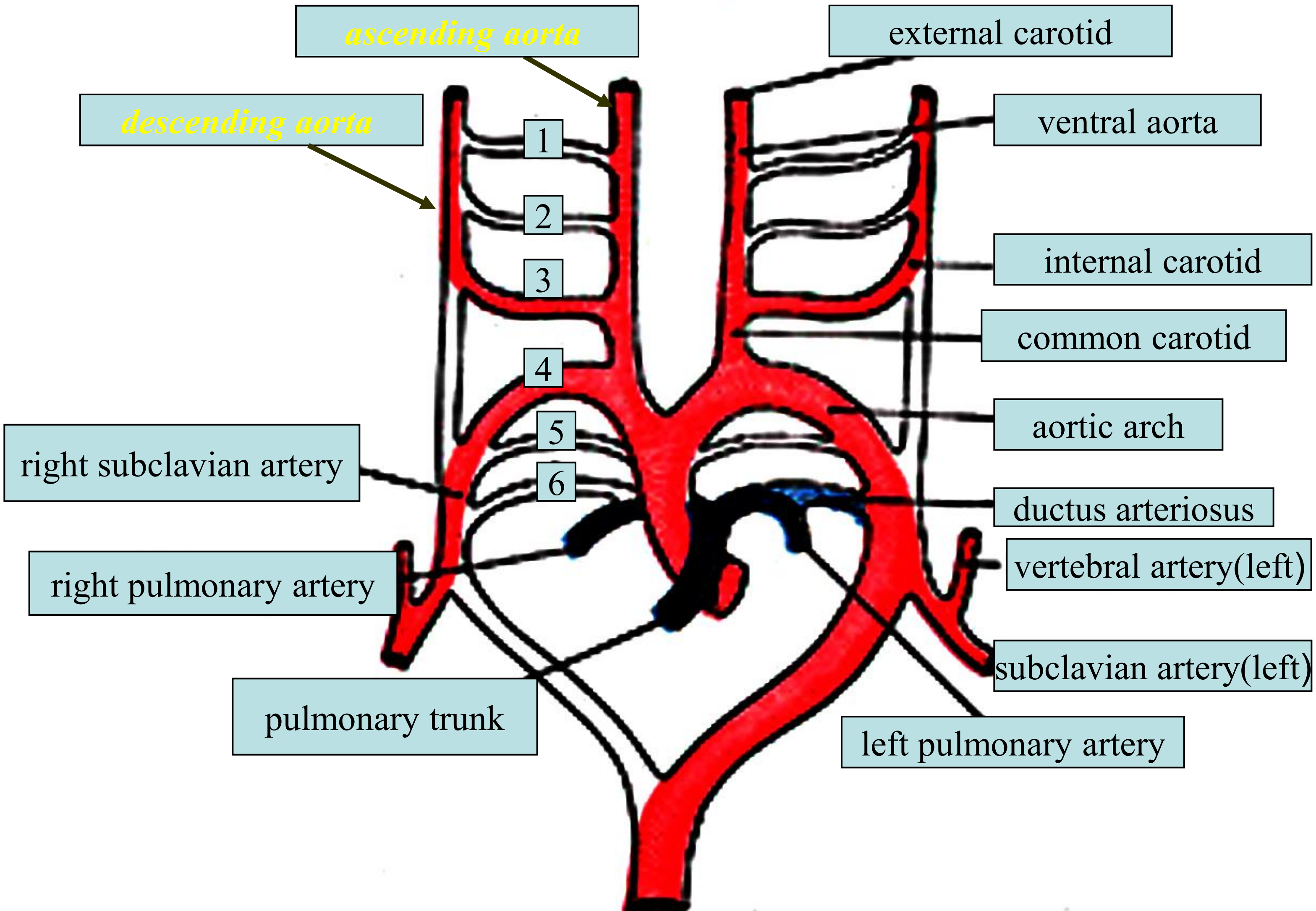
on the left side, the distal part persists as the **ductus arteriosus** during intrauterine life

the proximal part gives rise to the left branch of the pulmonary artery



The great arteries in the adult



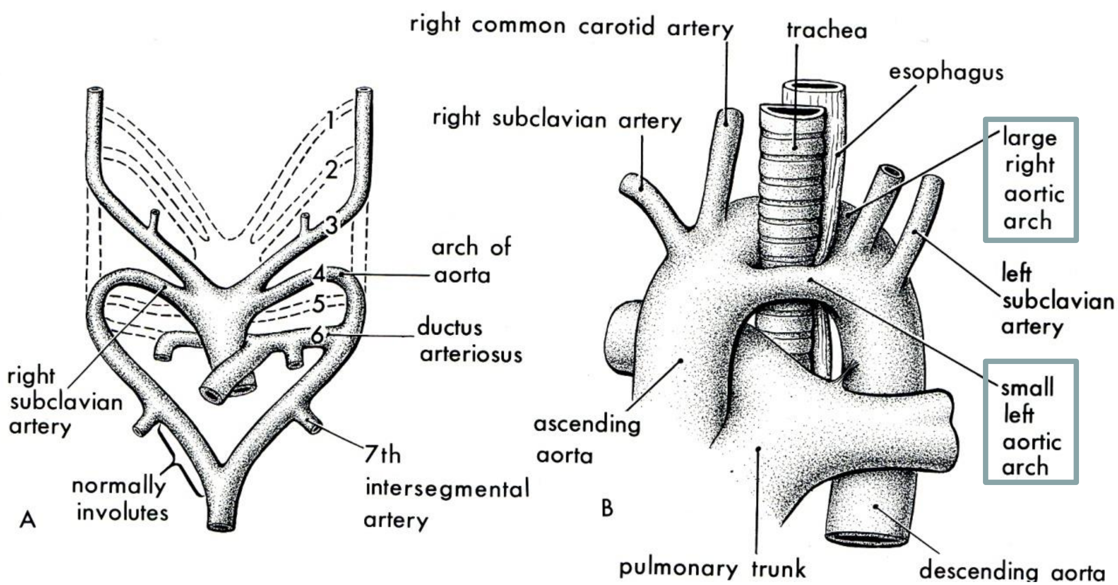


ascending aorta
external carotid
descending aorta
ventral aorta
1
2
3
internal carotid
common carotid
4
aortic arch
5
right subclavian artery
6
ductus arteriosus
vertebral artery(left)
right pulmonary artery
subclavian artery(left)
pulmonary trunk
left pulmonary artery

Anomalies of the aortic arches

- Double aortic arch:

  the distal portion of the right dorsal aorta persists and form right aortic



large *right* aortic arch and a small *left* one arise from the ascending aorta

↓

vascular ring around the trachea and esophagus

↓

!!There is a compression of the esophagus

Development of the venous system

At the 4th week 3 pairs of veins empty the *venous sinus (sinus venosus):*

- *Common cardinal veins:* from the embryo
- *Umbilical veins*: from the placenta
- *Vitelline (omphalomesentheric veins*: from the yolk sac

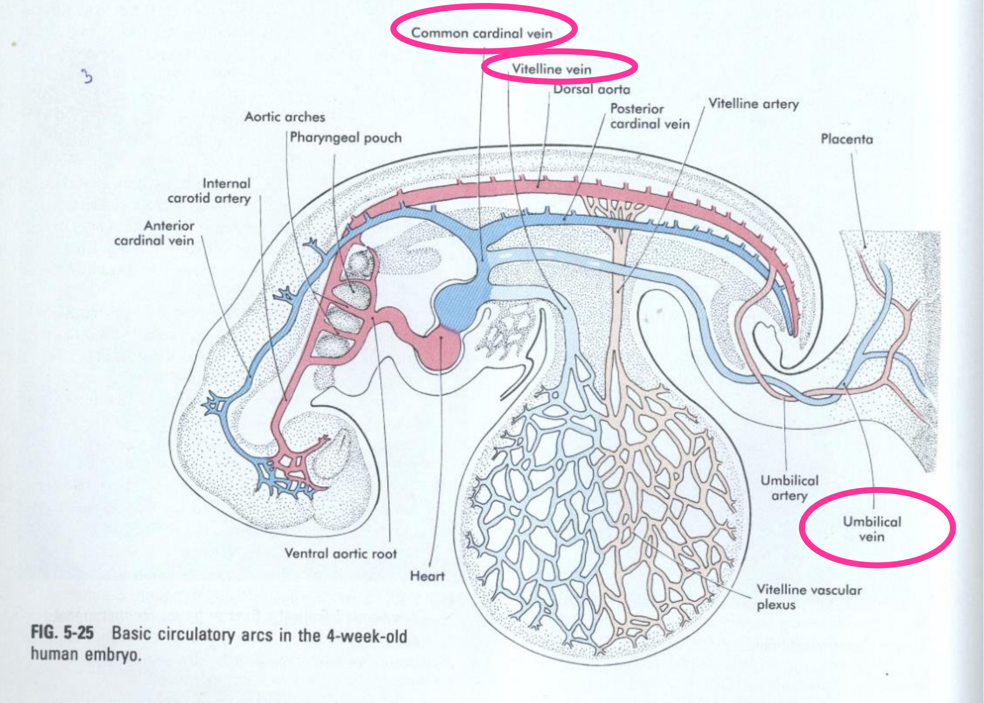


FIG. 5-25 Basic circulatory arcs in the 4-week-old human embryo.

Fetal circulation

Well oxygenated blood: from the *placenta* through the *umbilical vein*

hepatic sinusoids

ductus venosus (sphincter!)

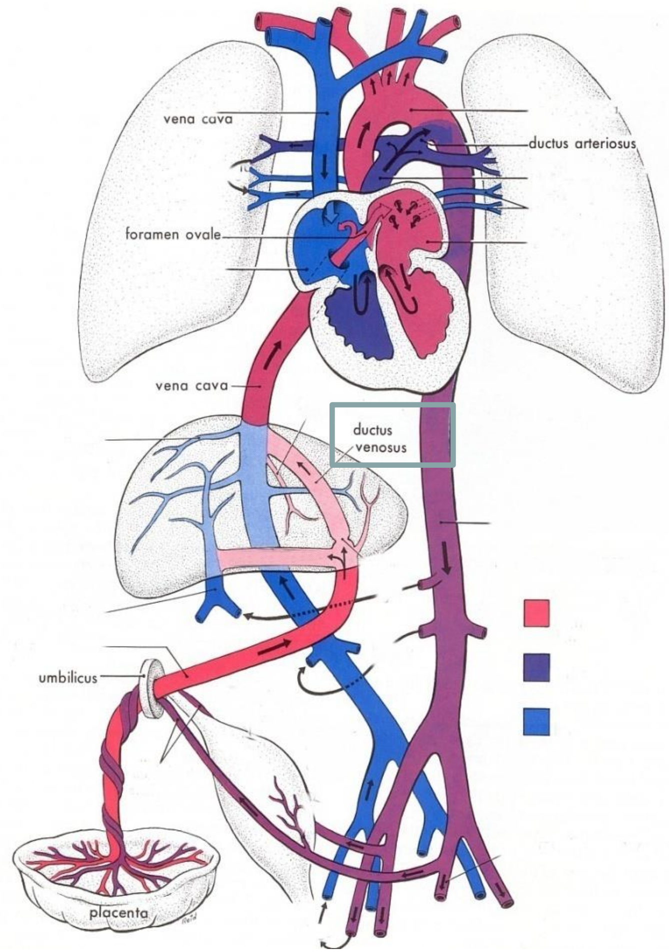
inf. v. cava

right atrium (blood is mixed: poorly oxygenated blood from the lower limbs, abdomen, pelvis, but still well-oixygenated)

pulmonary veins (small amount of deoxygenated blood) → *left atrium* (through foramen ovale)

left ventricle

ascending aorta



Fetal circulation

• small amount of well oxygenated blood from the inf. v. cava remains in the right atrium

↓

• mixes with poorly oxygenated blood from the sup. v. cava and coronary sinus

↓

right ventricle

↓

pulmonary trunk

some: to the lung

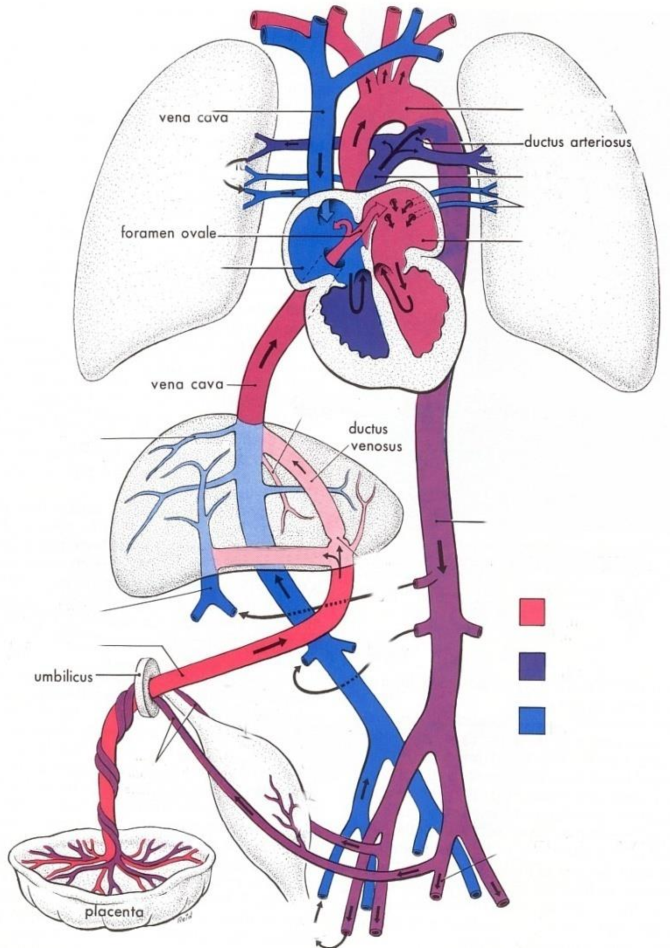
most: through the *ductus arteriosus to the aorta*

↓

umbilical arteries

↓

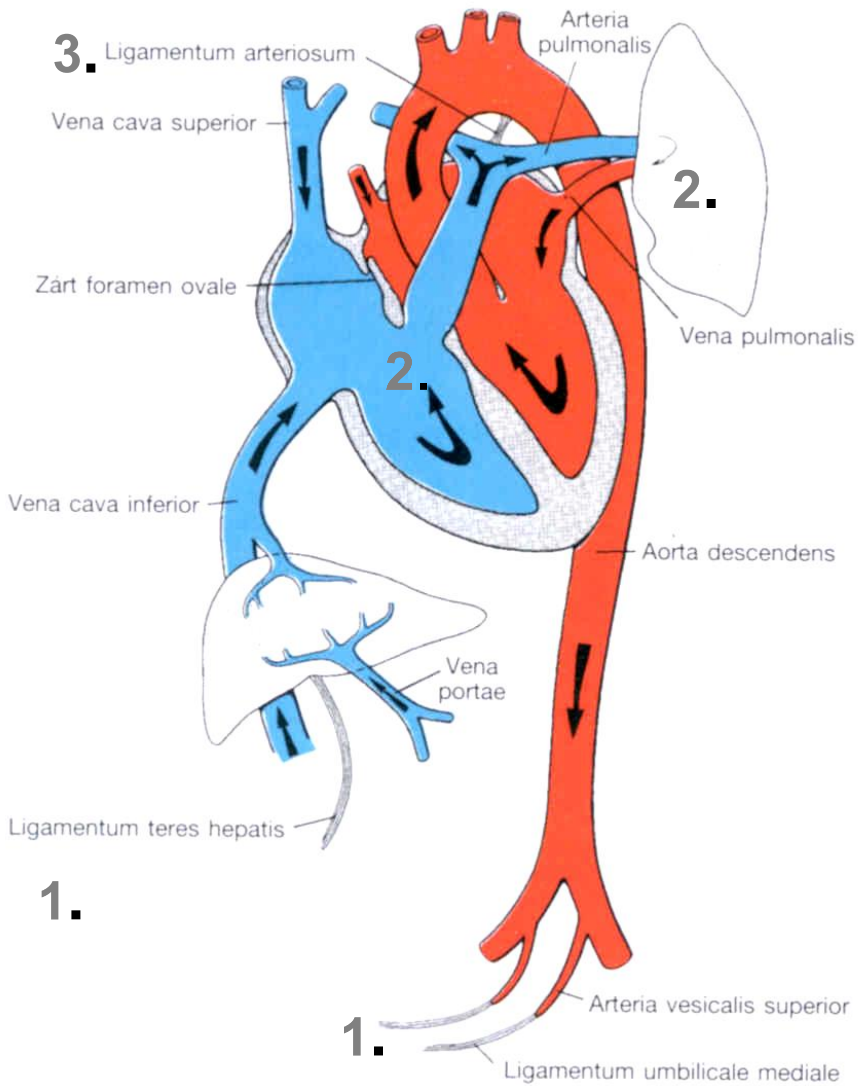
placenta

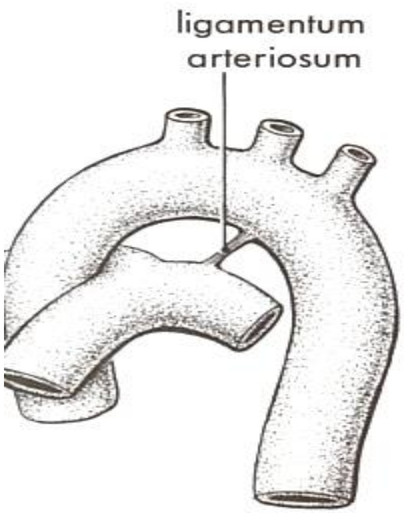


Perinatal changes

- **the infant lungs begin to function**
- **as soon as the baby is born foramen ovale, ductus arteriosus, ductus venosus umbilical vessels are no longer needed**

- the *sphincter* in the ductus venosus constricts ⟶ blood goes to the hepatic sinusoids
- occlusion of the placental circulation causes fall of blood pressure in the inf. v. cava
- as the lungs start to function, the vascular resistance in the pulmonary vessels dramatically decreases
- **pulmonary blood flow is increasing**
- **the pressure in the left atrium is raised**
- **this closes the foramen ovale by pressing the valve against the septum secundum**
- ductus arteriosus constricts at the birth ⟶ lig. arteriosum
- umbilical vessels constricts at the birth ⟶ lig. teres hepatis; plica umbilicalis medialis





Sandler T.W.:Langman’s Medical Embryolog